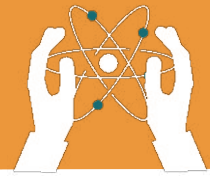


UNIT 5

VIRUSES, BACTERIA, FUNGI

Learning objectives:

- Understand the types of microorganisms: viruses, bacteria, and fungi.
- Recognize the helpful and harmful roles of microorganisms in everyday life.
- Explain safety precautions to prevent harmful effects from microorganisms.

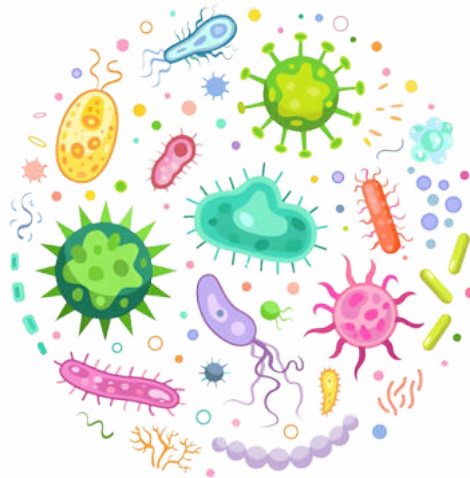


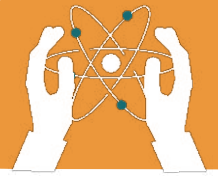
VIRUSES, BACTERIA, FUNGI

Microorganisms are minute unicellular organisms that are tiny living things that are too small to be seen with the naked eye. You need a microscope to see them. A person studying microorganisms in detail is known as **Microbiologist**. Microorganisms include viruses, bacteria, fungi, and other organisms like algae and protozoa. They are found everywhere in air, water, soil, and even inside our bodies.

What Are Microorganisms?

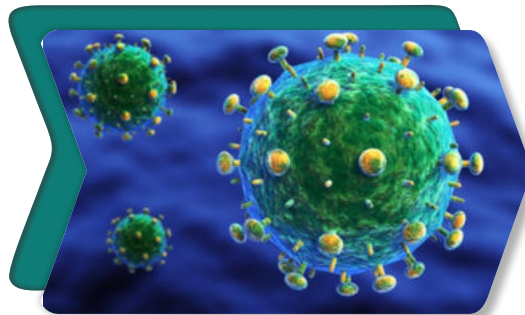
Microorganisms are tiny living things that can be helpful or harmful. Some help us, like bacteria that make yogurt, while others, like viruses, can make us sick.



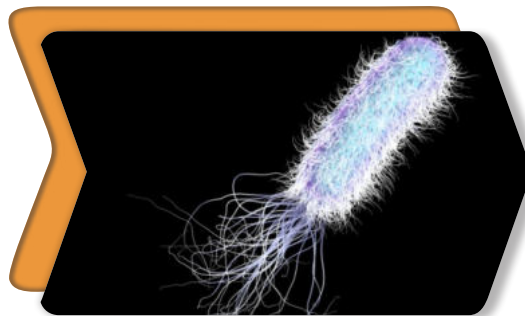


Types of Microorganisms

1. Viruses: Viruses: These are the smallest microorganisms. They come in different shapes, like rods, spheres, and even complex shapes like a space ship! Viruses can't live on their own; they need a living thing, called a host, to survive and reproduce. Viruses cause diseases like the flu, chickenpox, and COVID-19.

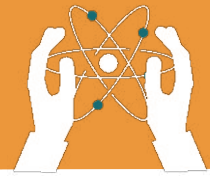


2. Bacteria: Bacteria are larger than viruses. They are found everywhere on our skin, in the air, and even in our stomach. Some bacteria are helpful, like those used in making cheese and yogurt, but others can cause infections, e.g. E. coli is a common gut bacteria.



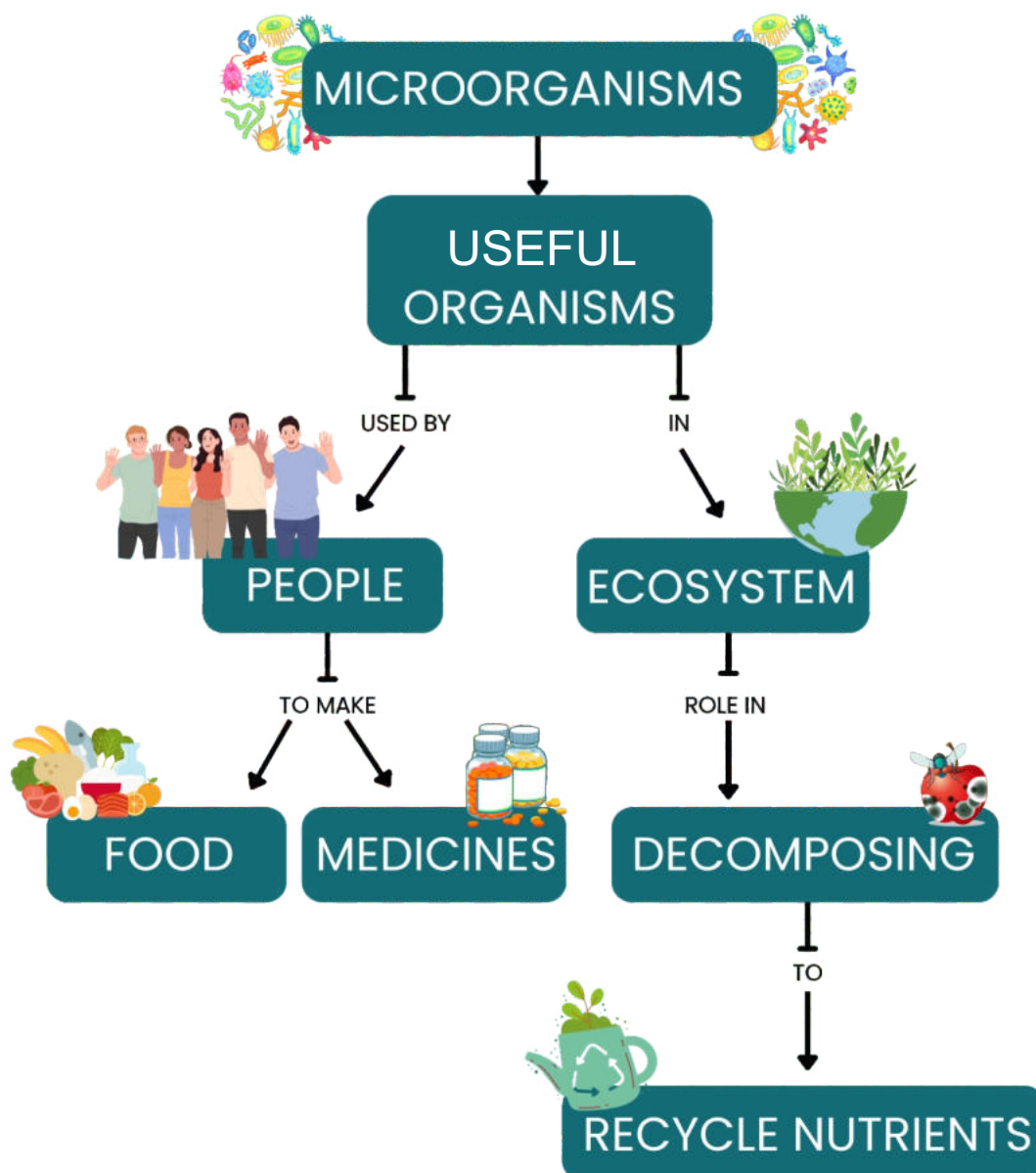
3. Fungi: Fungi include molds, and yeasts. Some fungi like mushrooms are not microorganisms. Some fungi are useful, like yeast, which makes bread fluffy, while others, like mold, spoil food.

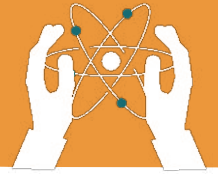




Role of Microorganisms in daily life:

- Bacteria help in making yogurt, cheese, and medicines like antibiotics.
- Yeast, a type of fungus, helps in baking bread and cakes.
- Harmless viruses are used to make vaccines that protect us from diseases.



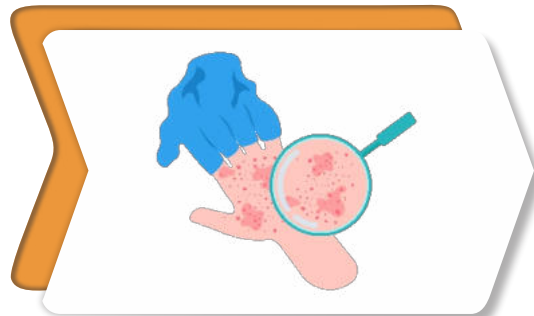


Harmful Roles of Microorganisms

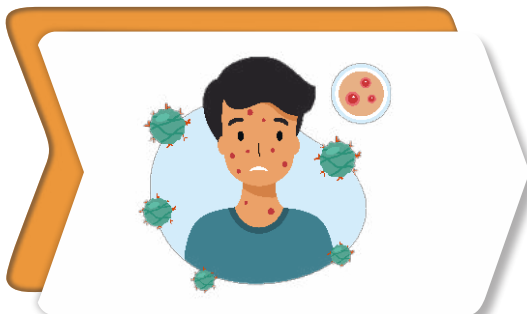
- Viruses cause illnesses such as the flu, measles, and COVID-19.
- Harmful bacteria can spoil food or cause diseases like sore throat and food poisoning.
- Fungi can cause skin infections like athlete's foot.



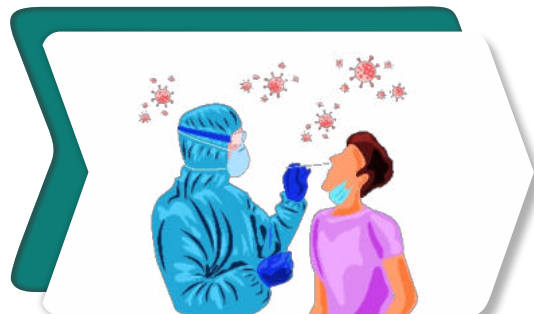
Contaminated Food



Skin disease athlete's foot



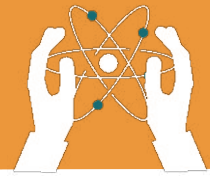
Symptoms of Illness



Medical Testing for Covid 19

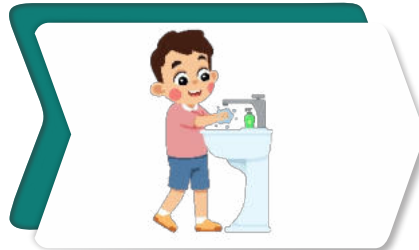


Recovery from flu



Safety and Precautions from Microorganisms

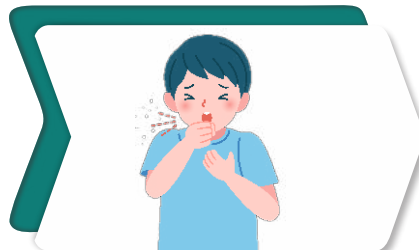
1.Wash Hands: Use soap and water regularly.



2.Cook and Store Food: Cook thoroughly and store properly



3.Cover Sneezes: Use a tissue while sneezing.

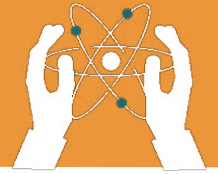


4.Clean Surroundings: Keep clothes and spaces tidy.



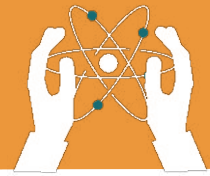
Get Vaccinated: Stay updated on vaccines.





Fill in the blanks.

1. Microorganisms are too small to be seen without a _____.
2. The person who studies microorganism in detail are known as _____.
3. _____ are the smallest microorganisms.
4. Some bacteria in our stomach help in _____ food.
5. _____ is a type of microorganism used to make bread rise.
6. Fungi can be helpful, but they can also cause _____.
7. Bacteria that spoil food are _____ to humans.
8. Viruses need a _____ to survive and reproduce.
9. _____ is a disease caused by a fungus.



Write “T” for True or “F” for False.

1. Viruses can survive without a host.

2. Bacteria are only harmful to humans.

3. Yeast is used to make bread fluffy.

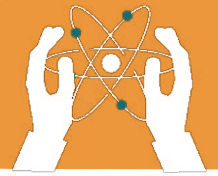
4. Some fungi are used in medicines.

5. Microorganisms are visible to the naked eye.

6. Bacteria help in making yogurt.

7. All microorganisms are harmful.

8. Fungi can grow on bread.



Choose the correct answer.

1. What is the smallest microorganism?

- Bacteria
 - Fungi
 - Virus
 - Algae
-

2. Which microorganism helps in making antibiotics?

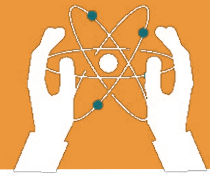
- Virus
 - Bacteria
 - Fungi
 - Protozoa
-

3. Which microorganism is used in making bread?

- Virus
 - Fungi
 - Bacteria
 - Algae
-

4. What does a virus need to live and reproduce?

- Air
- Host
- Water
- Light



5. Which microorganism can spoil food?

- Bacteria
 - Virus
 - Fungi
 - Protozoa
-

6. Where can microorganisms be found?

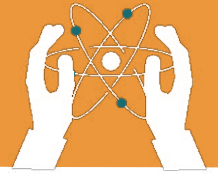
- Only in water
 - Everywhere
 - Only in air
 - Only in soil
-

7. What is athlete's foot caused by?

- Virus
 - Bacteria
 - Fungus
 - Algae
-

8. Which microorganism is used in making cheese?

- Bacteria
 - Virus
 - Fungi
 - Yeast
-



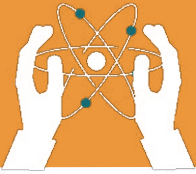
Answer the following questions.

- Q1. What are microorganisms, and where can they be found?
- Q2. How do bacteria help us?
- Q3. Name a harmful and a helpful role of fungi.
- Q4. How do viruses survive and reproduce?
- Q5. Why are some bacteria harmful?
- Q6. How is skin disease athlete's foot caused?
- Q7. Why is yeast important in baking?
- Q8. What can you do to protect yourself from harmful microorganisms?



Critical Thinking

What did you learn about mold growth?
Why does mold grow faster in warm
and wet places?



Grow Mold on Bread



ACTIVITY

- **Materials Needed:** A slice of bread, plastic bag and Some water.
- **Steps:**
 1. Take two slice of breads and note the changes occur in both slices.
 2. Take bread 1 and place it in open air without covering.
 3. Lightly sprinkle water over the slice of another bread.
 4. Place the bread inside a sealed plastic bag.
 5. Keep it in a warm spot for 3-5 days.
 6. Check the bread daily and observe changes. Write down your observations.

Fun Fact

- **Did you know?** Bacteria come in different shapes such as sphere, rods, chains etc.
- Viruses cannot be seen with ordinary microscope but they can be visible with electron microscope.
- There are more bacteria in your stomach than there are stars in the galaxy!